

Problem Statement

Ideation Phase: Problem definition and target users

Problem Statement

Project Title: FitBuddy — AI-Powered Personal Fitness Coach

Problem Definition

In today's fast-paced world, millions of people struggle to maintain a consistent fitness routine due to:

- **Lack of personalization:** Generic workout plans from the internet do not account for individual needs such as age, weight, fitness level, or specific goals.
 - **High cost of personal trainers:** Hiring a professional fitness coach is expensive and inaccessible to most people.
 - **Poor nutrition guidance:** People don't know what to eat to support their fitness goals, leading to poor results.
 - **No feedback loop:** Static plans can't adapt when users feel sore, busy, or want changes — causing dropout.
 - **Complexity:** Most fitness apps are overwhelming with too many features and require significant setup time.
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Problem Statement

"How might we build an AI-powered fitness application that generates fully personalized 7-day workout plans, adapts to user feedback in real-time, and provides actionable nutrition guidance — all through a simple and beautiful interface?"

Target Users

User Segment	Description
Beginners	People starting their fitness journey with no prior knowledge
Intermediate Users	People who exercise occasionally but lack structure
Busy Professionals	People with limited time who need efficient, short workouts
Goal-Focused Users	People targeting specific goals: weight loss, muscle gain, wellness

Pain Points Identified

- No personalization** — One-size-fits-all plans don't work.
- No adaptability** — Plans can't change based on daily feedback.
- Nutrition gap** — Exercise alone is insufficient without diet guidance.
- Expensive solutions** — Personal trainers and premium apps are costly.
- Drop-off rate** — Users lose motivation when plans are too rigid.

Solution Overview

FitBuddy solves these problems by:

- Collecting user profile data (name, age, weight, goal, intensity).
- Using **Google Gemini AI** to generate a unique 7-day workout plan.
- Allowing users to provide feedback and get the plan **updated in real-time**.
- Providing **personalized nutrition and recovery tips** based on the user's goal.
- Storing all data in a **SQLite database** for future reference and history tracking.

Expected Outcome

- A fully functional web application with an intuitive UI.
- AI-generated, personalized fitness plans in under 20 seconds.
- Users can refine their plan through natural-language feedback.
- Nutrition and hydration recommendations aligned with fitness goals.
- A scalable backend using FastAPI with persistent data storage.